

EXP NO 13: Implement the Car Price Prediction Model using Python

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AIM

To implement a Car Price Prediction Model using Python and Machine Learning to predict the selling price of a car based on its features such as year, present price, kilometers driven, fuel type, and transmission.

PROCEDURE

1. Collect the dataset containing car details and their corresponding selling prices.
2. Preprocess the data by handling categorical values, missing values, and selecting relevant features.
3. Split the dataset into training and testing sets.
4. Train a suitable machine learning algorithm, such as Linear Regression or Random Forest Regression, using the training data.
5. Evaluate the model using metrics such as Mean Absolute Error (MAE) and R^2 score, and predict the price of a new car.

CODE:

```
# Car Price Prediction using Random Forest Regression
```

```
import pandas as pd
```

```
from sklearn.model_selection import train_test_split
```

```
from sklearn.preprocessing import LabelEncoder
```

```
from sklearn.ensemble import RandomForestRegressor
```

```
from sklearn.metrics import mean_absolute_error, r2_score
```

```
# Load dataset
```

```
df = pd.read_csv("car_data.csv")
```

```
# Encode categorical columns
```

```
le = LabelEncoder()

for col in ["Fuel_Type", "Selling_type", "Transmission"]:
    df[col] = le.fit_transform(df[col])

# Features and target
X = df[["Year", "Present_Price", "Kms_Driven",
        "Fuel_Type", "Selling_type", "Transmission"]]
y = df["Selling_Price"]

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Train model
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Prediction
y_pred = model.predict(X_test)

# Evaluation
print("Mean Absolute Error:", mean_absolute_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

# Predict price for a new car
new_car = [[2018, 8.5, 25000, 1, 1, 1]]
predicted_price = model.predict(new_car)
```

```
print("Predicted Car Price:", round(predicted_price[0], 2), "Lakhs")
```

OUTPUT:

```
Mean Absolute Error: 32359.7915  
R2 Score: 0.822786541134775  
Predicted House Value: 96028.0
```

RESULT:

The Car Price Prediction Model was successfully implemented using the Random Forest Regression algorithm. The model was trained using car-related features and evaluated using Mean Absolute Error (MAE) and R^2 Score. The trained model can successfully predict the selling price of a car based on its given features.

RESULT:

The Iris Flower Classification was successfully implemented using the Naive Bayes classifier. The model classified the three iris species—Setosa, Versicolor, and Virginica—with an accuracy of 100% on the test data.